BoxCLIP: Vision–Language Joint Learning for Box-Supervised Remote Sensing Change Detection

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***Abstract*—Weakly supervised change detection (WSCD) aims to reduce annotation costs by replacing dense pixel-level labels with coarse-grained supervision. Within the WSCD paradigm, box-supervised change detection (CD) leverages bounding boxes to annotate change regions. However, existing box-supervised methods often struggle with inaccurate localization of subtle changes and blurred boundaries. To address these challenges, we propose BoxCLIP, a vision–language joint learning framework for box-supervised remote sensing CD (RSCD). The framework first generates image-specific land-cover semantic descriptions with large language models (LLMs), and fuses them with bitemporal visual features through a vision–language joint learning stage, thereby enhancing the semantic discriminability of change-aware representations. Guided by these LLM-derived semantic priors, the proposed method further applies level-set evolution under box-level supervision together with a joint self-supervised strategy, refining the coarse box annotations into precise physical contours and thereby alleviating boundary blurring while recovering fine-grained change instances. Extensive experiments on three public benchmark datasets, WHU-CD, CD-DATA-GZ, and LEVIR-CD, demonstrate that BoxCLIP achieves excellent performance under box-level supervision, attaining *F*1-scores of 92.16%, 85.02%, and 88.83%, respectively, while substantially reducing annotation costs and maintaining competitive efficiency. The code is available at https://github.com/TiffenYi/BoxCLIP**

***Index Terms*—Box-supervised learning, building change detection (BCD), large language model (LLM), remote sensing, weakly supervised learning.**

I. Introduction

HUMAN activities and natural evolutionary processes have profoundly transformed terrestrial landscapes. Multitemporal remote sensing imagery has emerged as a powerful tool for researchers to investigate land surface dynamics. As a fundamental technique in RS image analysis, change detection (CD) enables the localization of landscape modifications by systematically comparing diachronic images acquired at different temporal intervals within a specified period. In CD tasks, positive samples focus on capturing relevant "semantic changes," such as buildings or vehicles while filtering out disturbances like seasonal or lighting effects to ensure accurate detection [1]. In recent years, building CD (BCD) has become one of the most prominent and practically significant branches of the CD field, driven by its indispensable role in urban planning, land-use monitoring, disaster impact assessment, and cartographic map updating [2], [3].

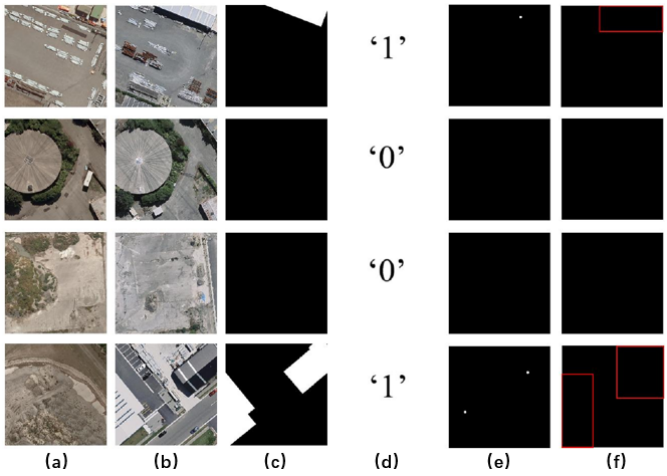


Fig. 1. Comparison of different annotation levels. (a) and (b) Images from different periods. (c) Pixel-level annotation. (d) Image-level annotation. (e) Point-level annotation. (f) Box-level annotation.

Prior to the widespread adoption of deep learning, most CD methods primarily focused on identifying changed pixels and classifying them to generate binary change maps. Classical machine learning models, such as support vector machines [4], *K*-means clustering, principal component analysis [5], decision trees [6], and conditional random fields [7], are commonly applied to CD tasks. Although these approaches demonstrate reasonable performance on certain datasets, they often struggle in complex scenes and large-scale RS imagery, showing limited accuracy and poor generalization.

The rapid development of deep learning has also brought about significant progress in the field of BCD. However, most of the existing CD methods rely heavily on precise pixel-level annotations, which require a significant amount of human

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and material resources. Some methods that use semisupervised learning have achieved good performance in related fields [8]. Recently, many researchers have shifted their focus to weakly supervised CD methods. In Fig. 1, different types of annotations are illustrated for comparison, including pixel-level annotation for fully supervised CD and image-level, point-level, and box-level annotations for weakly supervised CD. Pixel-level annotations contain the most complete semantic and positional information, enabling highly accurate detection, and are widely used in fully supervised CD [9], [10], [11]. Image-level annotations only indicate the category of objects in the image without providing spatial location, thereby significantly reducing labeling costs [12], [13]. Point-level annotations mark one or more key points within a target, offering a compromise in cost between image-level and box-level annotations [14], [15]. Box-level annotations, typically in the form of rectangular bounding boxes, provide both positional and category information. According to existing studies, labeling a pixel-level annotation takes approximately 15 times longer than a box-level annotation [9], [16]. Box-level annotations can also be generated automatically by extracting minimum bounding rectangles from change masks.

Several approaches employing different techniques have emerged in the weakly supervised CD (WSCD) field. AffinityNet [11] improves the accuracy of class activation maps (CAMs) by modeling differences between foreground and background regions, thereby delineating their boundaries and generating segmentation masks. Dai et al. [17] propose a CAM-based weakly supervised building CD method, which utilizes image-level supervision and introduces a multiscale joint supervision module and an improved consistency regularization module to enhance CAM quality. The multiscale joint supervision module improves model performance on changes of varying scales, while the improved consistency regularization module enhances boundary quality to generate high-quality predicted change maps. FCD-GAN [9] presents a region-based WSCD method that uses a segmenter to generate pseudochange masks, which are then applied to mask bitemporal images, prompting the segmenter to produce predictions that can deceive a discriminator. TransWCD [13] introduces both global and local priors, proposing two modules named dynamic prior decoder and label-gated constraint to alleviate missing change regions. MS-Former [18] adopts a transformer-based backbone and utilizes patch-level weakly supervised learning to extract differential features from image pairs. ACWCD [12] enhances WSCD by leveraging the internal attention mechanisms of vision transformers along with external prior knowledge.

Through an in-depth examination of this paradigm, we identify the primary challenge of WSCD as the lack of dense supervision, which manifests in blurred or imprecise instance boundaries and suboptimal localization of changed regions, a deficiency that is particularly pronounced for small-scale objects. In this context, the present study focuses on BCD, in which "semantic change" is defined as the appearance or disappearance of individual instances with well-defined boundaries.

To address these challenges, inspired by [19], [20], and [21], we propose BoxCLIP, a novel cross-modal weakly supervised BCD framework. Built on the CLIP architecture and adapted to the BCD task, the proposed method leverages large language models (LLMs) to analyze remote sensing imagery, thereby introducing category-level semantic priors, which are subsequently fused with image-based change features via a vision–language joint learning stage, where cross-attention facilitates effective integration for more precise localization of changed regions and improved overall performance. Moreover, by integrating a level-set algorithm with a joint self-supervised strategy within the change mask map generation stage, the proposed model refines instance boundaries under box-level supervision. Experiments on multiple remote-sensing CD benchmarks demonstrate that BoxCLIP substantially reduces reliance on pixel-level annotations while achieving significant gains in detection accuracy.

The main contributions of this article are as follows.

1) We propose a novel cross-modal box-supervised CD framework that leverages box-level annotations as supervision, along with semantic description texts for land-cover categories generated by LLMs to assist the detection process.
2) To fully exploit category semantic information for more accurate identification of change regions, we design a vision–language joint learning stage. By leveraging transformer-based dynamic alignment and cross-attention for cross-modal fusion, our framework effectively exploits textual prompts for semantic reasoning.
3) To better utilize the spatial cues of box-level labels, we build an adaptive boundary evolution stage that couples the adopted box-supervised level-set segmentation with a newly proposed orthogonal projection consistency loss and a joint self-supervised learning strategy, refining coarse boxes into precise boundaries for fine-grained CD.

The remainder of this article is organized as follows. Section II reviews the related work and discusses the motivation behind our approach. Section III details the proposed method and the architecture of each module. Section IV presents experimental results and performance evaluations on multiple benchmark datasets. Section V analyzes the model's performance and provides directions for parameter fine-tuning accordingly. Finally, Section VI concludes this article with a summary of our contributions and potential directions for future research.

II. Related Work

A. Fully Supervised CD

Fully supervised CD (FSCD) has achieved remarkable progress in recent years, benefiting from the strong feature representation capabilities of convolutional neural networks and transformers. Early works, such as FC-EF and FC-Siam-diff [22], established foundational CNN-based Siamese architectures that extract and compare bitemporal features. SNUNet [1] further improves spatial resolution preservation by introducing dense skip connections and efficient channel attention modules into the U-Net structure.

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To enhance contextual understanding, advanced CNN-based models incorporate attention and multiscale designs. RSCNet [10] introduces a self-calibrated convolution module that adaptively constructs spatial–spectral dependencies, improving feature discrimination. STANet [23] models spatiotemporal dependencies using dual attention branches and also contributes the widely used remote sensing CD (RSCD) benchmark. MIFNet [24] addresses the insufficient representation of global information and the low efficiency of multiscale feature fusion in CD by incorporating dual complementary attention modules and other mechanisms.

Recently, transformer-based methods have attracted increasing attention due to their global modeling capability. BiT [25] is one of the earliest attempts to apply vision transformers in CD tasks, enhancing spatial–temporal reasoning. DMATNet [26] introduces a dual-feature hybrid attention module and feedback mechanism, enabling finer spatial reconstruction. STeInFormer [27] proposes a spatial–temporal interaction transformer that combines multifrequency token mixing and interaction modules for improved efficiency and accuracy. CLAFA [28] improves CD accuracy by leveraging cross-level attention and learning attention residuals to refine features.

Although fully supervised CD methods offer high accuracy, they rely heavily on costly pixel-level annotations, motivating a growing interest in weakly supervised alternatives. Nonetheless, fully supervised frameworks remain the foundation of modern CD research and provide critical insights for designing more efficient, generalizable architectures.

B. Weakly Supervised CD

To mitigate the high cost of pixel-level annotations required by fully supervised methods, WSCD has gained increasing attention in recent years. These methods aim to perform CD using coarse-grained supervision signals such as image-level, region-level, or box-level labels while maintaining satisfactory detection performance.

Early work by Khan et al. [7] introduced a CRF-based WSCD framework, establishing a foundation for nonpixel-level supervision. Wang et al. [29] proposed SEAM, which combines self-supervision and consistency regularization to bridge the gap between weakly and fully supervised learning for semantic segmentation. Wu et al. [9] extended this idea in FCD-GAN, a GAN-based CD architecture capable of operating under various supervision levels, from unsupervised learning to fully supervised learning.

Recent WSCD methods primarily enhance feature discrimination and localization. Zhao et al. [30] proposed WSLCD with a dual-branch Siamese network and a refinement module for viewpoint invariance and spatial coherence. Han et al. [31] introduced HSANet, leveraging self-attention and cross-attention for multiscale semantics and boundary details, while Zhao et al. [32] presented DISep, an instance-separating framework that improves instance-level discrimination in dense high-resolution scenes.

Collectively, these methods demonstrate the potential of weakly supervised learning to approximate fully supervised performance, especially when enhanced with attention, contrastive learning, and region-level priors.

C. Vision–Language Models

The limitations of unimodal learning have spurred growing interest in text-based paradigms, as conventional models struggle to establish semantic correspondences between visual content and linguistic descriptions. Vision–language models address this gap by jointly understanding images (or video) and text and modeling their cross-modal associations. Typically, LLMs are pretrained on large-scale datasets using supervised learning [33] or self-supervised learning [34], and then adapted to downstream tasks following a "pretraining + fine-tuning" paradigm. For example, Deng et al. [35] employ transformers for referring expression grounding, aligning linguistic descriptions with specific image regions, while Rahhal et al. [36] develop a multilingual framework for cross-lingual image–text retrieval.

More recently, CLIP [37] has emerged as a seminal contribution to vision–language learning: by using contrastive learning to map images and text into a shared embedding space, CLIP enables powerful few-shot and zero-shot recognition. Trained on 400 million image–text pairs from the web—spanning diverse domains—CLIP demonstrates strong generalization across diverse downstream tasks. However, adapting CLIP to specific applications typically requires manually crafted prompts, which is expertise-intensive and time-consuming. To mitigate this limitation, context optimization (CoOp) [38] introduces learnable prompt vectors that facilitate task-specific adaptation without modifying CLIP's pretrained parameters, while Lin et al. [39] achieve state-of-the-art performance through linear classification in a unified representation space. In the RS domain, Dong et al. [21] introduce CLIP for fully supervised CD, leveraging paired image–text data to enable vision–language-driven change recognition.

Owing to its massive training dataset and powerful cross-modal representation capability, CLIP achieves excellent performance in various downstream tasks. Inspired by this success, we aim to incorporate language modality information into the encoding stage of weakly supervised CD, enhancing the semantic richness of visual features. This information enhancement enables more precise discrimination between changed and unchanged regions during decoding, thereby yielding more accurate change maps.

III. Methodology

In this section, we present a detailed description of the overall network architecture and the proposed modules. Section III-A provides an overview of the framework from a high-level perspective, including the input/output flow and the propagation of intermediate features within the network. Sections III-B–III-D introduce the core modules of the network and elaborate on their respective functionalities. Section III-E further explains the loss functions employed in our method.

A. Overview

The overall structure of our proposed method is illustrated in Fig. 2, which consists of three main stages: the semantic

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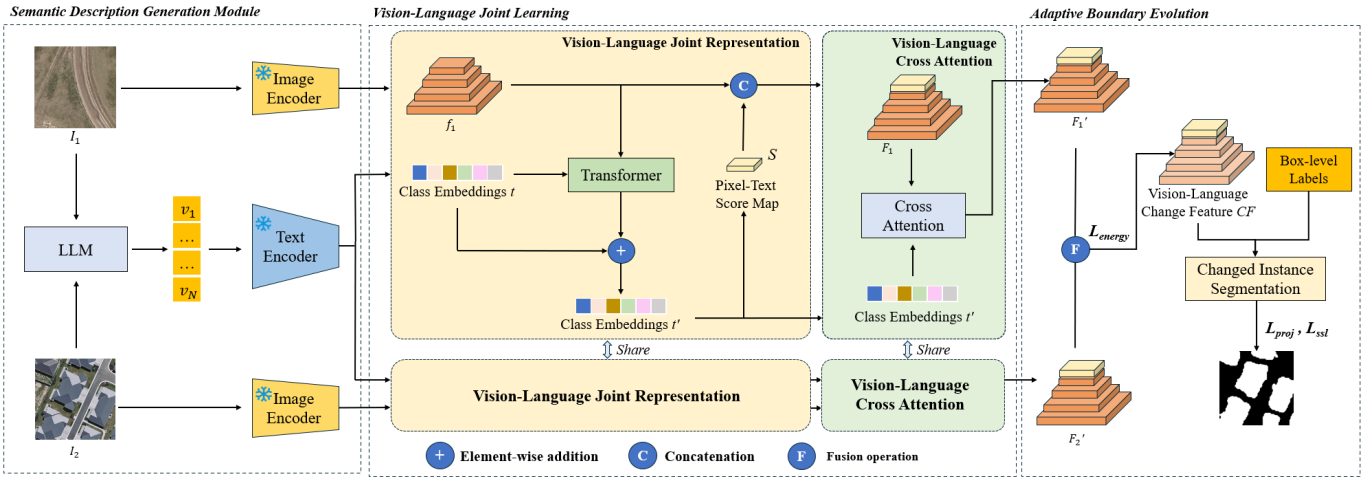


Fig. 2. **Overview of the proposed BoxCLIP framework.** The bitemporal images I_1 and I_2 are fed into an LLM, which generates image-specific land-cover semantic descriptions via a predefined semantic-constrained prompting strategy. A pretrained CLIP encoder then extracts multiscale visual features and text embeddings from the images and descriptions, respectively. The vision-language joint representation stage comprises VLJR and VLCA modules. In VLJR, a transformer-based design allows visual context to adaptively refine the textual representation, yielding a pixel–text score map. In VLCA, cross-attention is performed using image features as queries and text embeddings as keys and values, effectively integrating semantic knowledge from the language modality. For the adaptive boundary evolution stage, an FPN aggregates multiscale information, and a CIS head leverages box-level supervision. In addition, a self-supervised objective enforces consistency of features under affine transformations and promotes similarity across unchanged regions at different scales, producing the final change mask.

TABLE I
CATEGORY NAMES OF SOME COMMONLY ENCOUNTERED OBJECTS IN THE FIELD OF REMOTE SENSING

Road	Building	Farmland	Tree	Forest
Pond	Shrubbery	Impermeable Surface	Wetland	River
Lake	Parking Lot	Dense Residential	Fertile Land	Industrial Area
Sea	Freeway	Desert	Cars	Island
Mountain	Interchange	Runway	Solar Panel	Beach
Prairie	Airport	Meadow	Terrace	Storage Tanks
Harbor	Railway	Shoreline		

description generation stage, the vision–language joint learning stage, and the adaptive boundary evolution stage.

In WSCD, the fundamental challenge stems from the lack of dense supervision. To address this, the proposed approach resorts to vision–language alignment, where the core difficulty lies in bridging the modality gap between visual and textual representations. We initialize both the image and text encoders with CLIP [37] pretrained weights. A pair of bitemporal remote sensing images, I_1 and I_2 is input into the image encoder to extract feature maps f_1 and f_2. Meanwhile, a set of text prompts $\{v_i\}$ shown in Table I corresponding to the frequently occurring land-cover categories in the remote sensing building dataset is fed into the text encoder to obtain the initial class embeddings t. However, directly applying CLIP embeddings may introduce semantic inconsistency in domain-specific scenarios. Therefore, we refine t with guidance from the visual features to yield updated class embeddings t', which enhances the semantic adaptability to remote sensing change categories.

To address the lack of category priors and enhance the semantic discriminability of features, we compute a pixel–text score map using the updated class embeddings t' and concatenate it with the image features to obtain multimodal representations. These features are then refined through two submodules: the vision–language joint representation (VLJR) module and the vision–language cross-attention (VLCA) module. These components facilitate fine-grained cross-modal interactions and promote semantic alignment between vision and language modalities. Furthermore, to enable hierarchical spatial reasoning, a feature pyramid network (FPN) is used to aggregate multiscale features and produce fused change features CF.

Considering the problem of coarse or inaccurate boundaries in WSCD, we adopt the changed instance segmentation (CIS) method [40], which employs a kernel learning network to generate mask maps and leverages level-set evolution to improve the segmentation accuracy of box-level labels, ultimately producing smooth and shape-consistent instance-level change masks. In addition, we propose a joint self-supervised learning module that integrates consistency learning and a similarity loss. This dual-constraint mechanism jointly optimizes intermediate features and final predictions, thereby enhancing spatial boundary quality and improving the overall reliability of CD.

B. Semantic Description Generation

To transcend the inherent limitations of pixel-level comparisons, we introduce a semantic description generation stage designed to integrate high-level semantic information as auxiliary supervision for RSCD. Leveraging bitemporal imagery and a comprehensive classification taxonomy encompassing over 30 distinct land-cover categories, the framework generates precise, image-specific categorical textual prompts by

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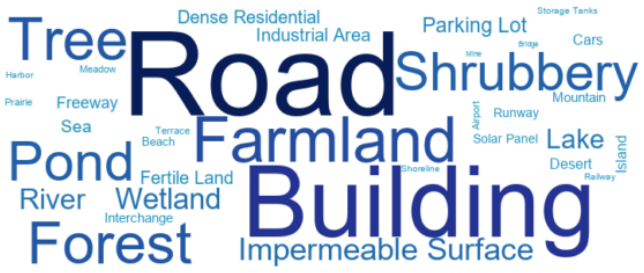


Fig. 3. Cloud map of the text prompt words.

providing constrained instructions to the Qwen3-VL large vision–language model. This approach moves beyond simple visual difference detection by incorporating explicit linguistic reasoning into the supervisory signal.

In particular, given a global template derived from the class label space $\mathcal{Y} \in \{1, 2, \ldots, N\}$, where N denotes the total number of categories. We query the model with a carefully designed structured prompt while constraining the generated land-cover categories to the scope predefined in Table I: "Within the predefined scope, list the N land-cover categories appearing in the remote sensing image, with each description following the format "an image of [class]." This process yields standardized linguistic descriptors, which are subsequently encoded through a frozen CLIP text encoder to generate the textual embedding. The cloud map of the text prompt words for the category of ground objects in remote sensing images generated by LLMs is shown in Fig. 3. These semantic cues are then integrated via a dual-branch architecture, effectively injecting rich prior knowledge into the learning process. Within this integrated framework, a shared latent embedding space is established to bridge the gap between high-level visual representations and categorical linguistic embeddings. By leveraging such textual information as abstract supervision, the mechanism guides the backbone network toward semantic-aware representation learning. Consequently, this approach fosters robust cross-modal associations and mitigates the dependence on labor-intensive pixel-level annotations. The detailed operations are defined as follows:

$$f_i = \text{ImgEncoder}(I_i), \quad i \in \{1, 2\} \tag{1}$$

$$t_i = \text{TextEncoder}(V_i), \quad t_i \in \mathbb{R}^{n \times d} \tag{2}$$

where f_i denotes the feature map extracted from the input image I_i using CLIP's image encoder. t_i represents the class-level text embeddings, where $V_i = \{v_1, v_2, \ldots, v_n\}$ denotes the set of n prompt word vectors, and d is the dimension of the text embeddings, which matches the dimension of f_i.

C. Vision–Language Joint Learning

In the vision–language joint learning stage, we propose two modules: the VLJR module and the VLCA module. These modules enable effective fusion of image and text features, thereby enhancing the semantic representation power of the visual features.

1) Vision–Language Joint Representation: From the previous stage, we obtain the image features f_i and the class-level text embeddings t. These features are fed into the VLJR module to generate the pixel–text score map s. The process is defined as

$$\bar{f}_i = GAP(f_i) \tag{3}$$

$$[\bar{z}_i, z_i] = MHSA\left(\bar{f}_i, f_i\right), \quad \bar{z} \in \mathbb{R}^{1 \times C}, \ z \in \mathbb{R}^{H \times W \times C} \tag{4}$$

$$t_i' = t_i + \gamma \cdot Trans(t_i, [\bar{z}_i, z_i]) \tag{5}$$

$$s_i = \hat{z}_i \cdot \hat{t}_i', s_i \in \mathbb{R}^{H \times W \times K} \tag{6}$$

$$F_i = Concat(f_i, s_i) \tag{7}$$

where $\bar{f}_i$ represents the global contextual feature obtained by applying the global average pooling operation, denoted as $GAP(\cdot)$. $MHSA(\cdot, \cdot)$ refers to the multihead self-attention mechanism. $Trans(\cdot, \cdot)$ is the transformer decoder. $Concat(\cdot, \cdot)$ indicates concatenation along the feature dimension. The refined text embedding t' is obtained by integrating the original text embedding t with the output of the transformer decoder, scaled by a learnable parameter γ, thereby enabling adaptive incorporation of visual context into the textual representation. $\hat{z}$ and $\hat{t}'$ represent the L_2-normalized versions of z and t'. s is the resulting pixel–text score map. Finally, we concatenate s and f_i to obtain the joint feature.

2) Vision–Language Cross-Attention: To enable visual features to attend to semantic cues from the text embeddings, we employ a cross-attention mechanism in the VLCA stage. The attention is computed with h heads. Multihead attention is realized by projecting the queries (Q), keys (K), and values (V) into distinct subspaces via separate learnable linear projections; attention is evaluated independently in each head, and the results are concatenated and linearly transformed. The query is derived from the feature maps F, which are the output of the VLJR module, and the key and value are derived from the refined text embedding t'. The process is defined as follows:

$$F^v = Reshape(F + Pos) \tag{8}$$

$$Q = F^v W_n^Q, \quad K = t' W_n^K, \ V = t' W_n^V \tag{9}$$

where $F^v \in \mathbb{R}^{HW \times C}$ is the reshaped form of $F \in \mathbb{R}^{H \times W \times C}$ after adding the positional embedding $Pos \in \mathbb{R}^{H \times W \times C}$, $n \in [1, h]$ represents the index of the attention head, and $W_n^Q \in \mathbb{R}^{C \times d}$, $W_n^K, W_n^V \in \mathbb{R}^{D \times d}$ are learnable projection matrices. The cross-attention operation is performed as

$$\begin{aligned} head_n &= SingleHead(Q, K, V) \\ &= Softmax\left(\frac{QK^T}{\sqrt{d}}\right) V \end{aligned} \tag{10}$$

$$M = Concat(head_n) W^M \tag{11}$$

$$F' = CA\left(F, t'\right) = LN\left(F + ReLU\left(MW^{L1}\right) W^{L2}\right) \tag{12}$$

where $Softmax(\cdot)$ is the softmax function and $SingleHead(\cdot)$ is single-head self-attention operation. M represents the output of concatenation over attention heads. $W^M \in \mathbb{R}^{hd \times C}$, $W^{L1} \in \mathbb{R}^{C \times 4C}$, and $W^{L2} \in \mathbb{R}^{4C \times C}$ are trainable weights. $ReLU(\cdot)$ denotes the activation function, and $LN(\cdot)$ denotes layer normalization. Through the cross-attention mechanism $CA(\cdot, \cdot)$, visual-language features F' can be ultimately obtained.

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This stage allows the visual-language features to effectively integrate semantic knowledge from the language modality, further enhancing the discriminative power of the joint representation.

D. Adaptive Boundary Evolution

After obtaining the vision–language features of the bitemporal images through the dual-branch network with shared weights, we will extract their difference information through a fusion module. The specific implementation of the fusion operation is shown below

$$Fuse\left(F_1', F_2'\right) = Conv\left(Concat\left(F_1', F_2'\right)\right) \tag{13}$$

$$CF = FPN\left(Fuse\left(Sub\left(F_1', F_2'\right), Fuse\left(F_1', F_2'\right)\right)\right) \tag{14}$$

where $Fuse(\cdot,\cdot)$ denotes the fusion operation on the extracted bitemporal feature. $Conv(\cdot)$ is a 1×1 convolution used to reduce the feature dimension. $Sub(\cdot,\cdot)$ is the operation of subtracting and taking the absolute value. $FPN(\cdot)$ is used to unify the dimensions of multiscale features and multiscale feature fusion, to improve the detection ability of small objects. CF is the vision–language changed feature.

1) Changed Instance Segmentation: Inspired by [41], we introduce a level-set energy functional into the optimization procedure and optimize it via gradient backpropagation. Given a level-set function $\phi(x,y)$ that partitions the image domain Ω into foreground and background separated by the zero-level contour $c = \{(x,y) : \phi(x,y) = 0\}$, the energy functional $\mathcal{L}(.,.,.)$ restricted to the region inside the box label B is defined as

$$\begin{aligned}\mathcal{L}(\phi, I, B) = &\int_{\Omega\in B} |I(x,y) - c_1|^2\ \sigma(\phi(x,y))\ dx\,dy \\ &+ \int_{\Omega\in B} |I(x,y) - c_2|^2 (1 - \sigma(\phi(x,y)))\ dx\,dy \\ &+ \gamma \int_{\Omega\in B} |\nabla\sigma(\phi(x,y))|\ dx\,dy\end{aligned} \tag{15}$$

$$c_1 = \frac{\displaystyle\int_{\Omega\in B} I(x,y)\ \sigma(\phi(x,y))\ dx\,dy}{\displaystyle\int_{\Omega\in B} \sigma(\phi(x,y))\ dx\,dy} \tag{16}$$

$$c_2 = \frac{\displaystyle\int_{\Omega\in B} I(x,y)\ \ (1 - \sigma(\phi(x,y)))\ dx\,dy}{\displaystyle\int_{\Omega\in B} (1 - \sigma(\phi(x,y)))\ dx\,dy} \tag{17}$$

where $\sigma(\cdot)$ denotes the sigmoid function, γ is a weighting parameter, and B is the box-level label. The region means c_1 and c_2 are the average intensities of the normalized input $I(x,y)$ over the foreground and background.

Minimizing the energy functional in (15) drives the model to iteratively evolve the coarse box annotations into precise physical contours, which yields the adaptive contour evolution loss L_{energy}. Since only positive change instances contribute to supervision, we introduce a masking operator that retains a predicted change mask M only when it is a positive sample and discards it otherwise

$$\text{Condition}(M) = \begin{cases} M, & \text{if } M > 0 \\ \phi, & \text{otherwise} \end{cases} \tag{18}$$

$$L_{\text{energy}} = \frac{1}{V} \sum_j \sum_{k=1}^{4} \text{Condition}\left(\mathcal{L}\left(\phi_j, M^k, B\right)\right) \tag{19}$$

where V is the number of positive change-instance masks, M^k denotes the output of the kth level of the feature pyramid ($k \in \{1,2,3,4\}$), which embeds richer semantic information for more robust results, and ϕ_j is the boundary curve of the jth positive sample. To fully exploit pixel- and spatial-affinity information while preserving the local consistency of the level-set predictions, we apply this operation to the predicted level set at each training step for each positive sample j.

In addition, an orthogonal projection consistency loss L_proj is introduced to enforce agreement between the predicted masks and the box-level labels. It first projects the 2-D predicted mask P onto the horizontal (X) and vertical (Y) axes, yielding the 1-D vectors

$$p_x = \max_y P, \quad p_y = \max_x P \tag{20}$$

$$\text{Dice}(p,b) = 1 - \frac{2\sum_{i=1}^{N} p_i b_i + \epsilon}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} b_i + \epsilon} \tag{21}$$

$$L_{\text{proj}} = \frac{1}{V} \sum_j \text{Condition}\left(\text{Dice}(p_x, b_x) + \text{Dice}\left(p_y, b_y\right)\right) \tag{22}$$

where $\max_x$ and $\max_y$ denote the maximum operations along the X- and Y-axes, respectively; p_x and p_y are the projections of the predicted change mask onto the X- and Y-axes; b_x and b_y are the corresponding projections of the box-level label; and $\text{Dice}(\cdot,\cdot)$ computes the Dice coefficient defined in (21).

Under the box-level supervision, the adaptive contour evolution loss L_{energy} and the orthogonal projection consistency loss L_{proj} jointly drive the coarse box annotations to iteratively evolve toward accurate physical contours, thereby alleviating boundary blurring, incomplete change regions, and small-object omission that typically arise from the lack of fine-grained pixel-level supervision in weakly supervised remote sensing image CD.

2) Joint Self-Supervised Learning: Self-supervised learning has drawn increasing attention from researchers. It improves the self-learning ability of the model in a low-cost way, encourages the model to self-discover patterns in the data, and has shown great potential in computer vision, natural language processing, and other fields [34], [42]. To enhance the model's adaptability and boundary prediction quality, we introduce the joint self-supervised learning stage, which integrates two self-supervised learning mechanisms. The underlying principles of these components are detailed in this section.

1) *Self-Supervised Consistency Learning:* The basic idea of this module is that the results generated by the feature after affine transformation should be the same as the results generated by the original feature after the same transformation operation [41]. As shown in [43], consistency regularization can be used in the feature maps instead of being used in the original image. The affine transformation is essentially a superposition of a linear transformation and a translation transformation, and it introduces no additional changes.

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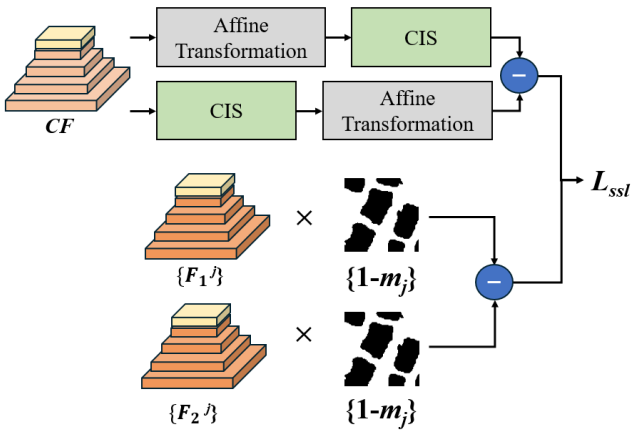


Fig. 4. **Illustration of joint self-supervised learning.** Joint self-supervised learning improves the robustness of the model and the boundary quality of the model. It consists of SSCL and SSSL.

2) *Self-Supervised Similarity Learning:* A self-supervised similarity learning (SSSL) module is designed to improve boundary quality. It measures the feature similarity between the two temporal phases, encouraging the features of the two phases to remain similar outside the changed regions while allowing them to differ within the changed regions; in other words, it enforces that the regions not belonging to changes (i.e., the unchanged regions) preserve consistent semantics between the two phases. Since the unchanged regions should be highly similar, the constraint is imposed on the unchanged regions only, by masking the jth stage features F_1^j and F_2^j with $(1 - m_j)$, where m_j is the binary box-level mask. In the final configuration, only this unchanged-region similarity term is activated in L_{ssl}. As shown in Fig. 4, this method integrates the aforementioned two self-supervised modules and designs a unified self-supervised loss function L_{ssl}, whose specific formulation is presented as follows:

$$m' = T\left(CIS\left(CF\right)\right), \quad m'' = CIS\left(T\left(CF\right)\right) \tag{23}$$

$$L_{ssl} = \left|\left|m' - m''\right|\right| + \sum_j \left|\left|F_1^j * \left(1 - m_j\right) - F_2^j * \left(1 - m_j\right)\right|\right| \tag{24}$$

where $CIS(\cdot)$ refers to the above CIS module. $T(\cdot)$ refers to the affine transformation. CF is the vision–language changed features. $\| * \|$ denotes the L_2 norm. Here m' and m'' refer to the output from different sequences of affine transformation and CIS operation. Together with the consistency learning, this improves boundary quality and overall reliability.

E. Network Details and Losses

1) Network Details: The designed network adopts a Siamese architecture based on the pretrained CLIP vision transformer (ViT-B/16), in which two weight-sharing branches extract multiscale features from bitemporal images. For an input of size $H \times W$, the backbone outputs four feature-map pairs with resolutions of $H/4$, $H/8$, $H/16$, and $H/32$, each of 768 channels. The frozen pretrained text encoder encodes the K class-level prompts, from which a K-channel pixel-to-text score map is computed. This map is then concatenated with the third-level features, resulting in $768 + K$ channels. The VLCA module then fuses the text semantics into every level via cross-attention without changing its channel number. Per level, bitemporal features are fused by channel concatenation, absolute subtraction, and 1×1 convolution, yielding 256-channel change features that are aggregated by a FPN into five scales at strides of 4–64. Finally, the generated masks are further refined under box-level supervision by means of level-set evolution combined with a self-supervised strategy.

2) Losses: We optimize the network with three terms: level-set energy loss L_{energy} on positive masks to encourage smooth and shape-consistent boundaries, L_{proj} for instance segmentation with box annotations, and L_{ssl} combines two self-supervised strategies: consistency of features under affine transformations and similarity of unchanged regions across different scales. The loss function L_{total} to train the BoxCLIP network consists of the above three items

$$L_{total} = \lambda_1 L_{energy} + \lambda_2 L_{proj} + \lambda_3 L_{ssl} \tag{25}$$

where λ_1, λ_2, and λ_3 are three tradeoff parameters.

IV. Experiments

A. Dataset Description

Due to the current absence of dedicated datasets for box-level CD, we propose to leverage existing public CD datasets with pixel-level annotations to construct the required benchmark. Our methodology for generating box-level CD annotations involves extracting connected regions from pixel-wise ground truth and subsequently delineating their minimum enclosing bounding boxes as instance-level detection targets. The proposed approach is systematically validated on three established public datasets widely recognized in the CD research community: The WHU-CD dataset [44], the LEVIR-CD dataset [23], and CD-DATA-GZ [45] dataset.

1) *WHU-CD [44]:* The WHU-CD dataset contains aerial images of Christchurch, New Zealand, which have a very high spatial resolution (0.075 m/pixel) and cover an area of 450 km^2. A total of more than 220 000 individual buildings were sampled in the dataset, which were extracted from the aerial images at a spatial resolution of 0.075 m.
2) *LEVIR-CD [23]:* The LEVIR-CD dataset is a large-scale remote sensing BCD dataset provided by the LEVIR laboratory of Beijing University of Aeronautics and Astronautics. The dataset contains 637 pairs of very high-resolution Google Earth image patches (VHR, 0.5 m/pixel), each of which is 1024×1024 pixels.
3) *CD-DATA-GZ [45]:* CD-DATA-GZ is a high-resolution remotely sensed CD dataset, which collects VHR image pairs in the suburbs of Guangzhou City, China, from 2006 to 2019. The Google Earth service with BIGEMAP software was used to collect 19 pairs of seasonally varying VHR images with red, green, and blue bands,

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a spatial resolution of 0.55 m, and sizes ranging from 1006×1168 pixels to 4936×5224 pixels.

To accelerate processing, the images of each dataset are first partitioned into training, validation, and test sets at the original image level, following the official split protocol of each benchmark, and are subsequently cropped into 256×256 patches. Since all patches cropped from the same source image are confined to a single set, spatially adjacent patches never cross set boundaries, thereby preventing data leakage between training and testing.

B. Implementation Details and Evaluation Metrics

1) Implementation Details: All experiments were carried out using the open-source PyTorch framework and the MMDetection toolbox on an NVIDIA RTX 3090 GPU with 24 GB of memory. All image pairs were uniformly resized to 256×256 pixels and augmented with random cropping and horizontal flipping to increase the diversity of training samples. The proposed network was trained for 96 epochs with the AdamW optimizer, using an initial learning rate of 1×10^{-4}, a weight decay of 1×10^{-4}, and a batch size of 8; gradient clipping with a maximum norm of 1.0 was adopted to ensure stable convergence. Our framework is built upon the pretrained CLIP model (ViT-B/16), which consists of three components, namely the image backbone, the text encoder, and the context decoder. To preserve the potential connection between visual and textual features in the pretrained model as much as possible, the backbone learning rate was set to one-tenth of the overall learning rate (i.e., 1×10^{-5}), and the text encoder was kept frozen throughout training so that its pretraining weights remained unchanged, thereby ensuring that the model can effectively exploit the pretrained knowledge. For the text modality, the Qwen3-VL large vision–language model is first employed to process the BCD dataset as described in Section III-B; the resulting text embeddings are then fed into the network and participate in training together with the corresponding bitemporal images.

2) Evaluation Metrics: In our experiments, five widely used metrics, precision (Prec), recall (Rec), $F1$ score ($F1$), IoU, and Boundary IoU (BIoU) [46], are employed to evaluate the performance of the CD task. The detailed calculations of the four metrics are as follows:

$$Precision = \frac{TP}{TP + FP} \tag{26}$$

$$Recall = \frac{TP}{TP + FN} \tag{27}$$

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall} \tag{28}$$

$$IoU = \frac{TP}{TP + FN + FP} \tag{29}$$

$$BIoU = \frac{|(G_d \cap G) \cap (P_d \cap P)|}{|(G_d \cap G) \cup (P_d \cap P)|} \tag{30}$$

where G denotes the ground-truth mask, P denotes the predicted mask, and G_d and P_d are the masks obtained by dilating G and P by a distance d.

C. Comparison Results With WSCD Methods

We compared the proposed model with five state-of-the-art weakly supervised RSCD methods, including FCD-GAN [9], TransWCD [13], MJS-ICR-Net [17], MS-Former [18], and ACWCD [12]. A brief introduction to the comparative methods is provided as follows.

1) *FCD-GAN [9]:* A weakly supervised model based on generative adversarial networks, which learns to generate accurate change masks using only image-level supervision without pixel-level annotations.
2) *TransWCD [13]:* A transformer-based weakly supervised approach that effectively exploits both global and local priors for WSCD, achieving state-of-the-art performance on several benchmark datasets.
3) *MJS-ICR-Net [17]:* This method integrates an MJS module and an ICR module, where the MJS module facilitates multiscale change feature extraction through the generation of refined CAMs, and the ICR module enhances the consistency of these CAMs to improve the precision of edge feature representation.
4) *MS-Former [18]:* This method is a transformer-based network for WSCD. Differential features are extracted from patch-level annotations using attention mechanisms, thereby better distinguishing changed and unchanged regions.
5) *ACWCD [12]:* This method leverages the inherent transformer information and prior knowledge to perform CD based on image-level annotations, thereby improving the quality of pseudolabel generation.

The MJS-ICR-Net was reimplemented by us and executed under the same experimental settings as the proposed method. FCD-GAN,[1] TransWCD,[2] MS-Former,[3] and ACWCD[4] are implemented directly using the code framework provided by the authors.

1) WHU-CD Dataset: As reported in Table II, our method attains the best performance on the WHU-CD dataset across all evaluation metrics, achieving an $F1$-score of 92.16%, an IoU of 85.46%, and a BIoU of 95.31%. Relative to ACWCD ($F1$: 78.73%, IoU: 64.92%), our method improves $F1$ and IoU by 13.43% and 20.54%, respectively, and elevates the BIoU from 87.99% to 95.31%, reflecting superior boundary fidelity. The high recall of 94.92%, combined with a precision of 89.56%, indicates that our method effectively suppresses missed detections while maintaining false positives at a moderate level. On the contrary, FCD-GAN exhibits an extremely low precision of 9.05%, signifying severe false-positive errors and weak discriminative capability, whereas MS-Former suffers from a low recall of 57.32%, indicating incomplete change localization.

Fig. 5 presents qualitative comparisons on the WHU-CD dataset, where five image pairs covering both small-scale and large-scale building changes are selected for evaluation. For small-scale building changes, our method markedly

[1]FCD-GAN: https://github.com/Cwuwuhu/FCD-GAN-pytorch
[2]TransWCD: https://github.com/zhenghuizhao/TransWCD
[3]MS-Former: https://github.com/guanyuezhen/MS-Former
[4]ACWCD: https://github.com/WenhaoLiu03/ACWCD

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TABLE II
EVALUATION METRICS OF WSCD ON THREE BENCHMARK DATASETS

Methods	WHU-CD					CD-DATA-GZ					LEVIR-CD				
	Prec(%)	Rec(%)	F1(%)	IoU(%)	BIoU(%)	Prec(%)	Rec(%)	F1(%)	IoU(%)	BIoU(%)	Prec(%)	Rec(%)	F1(%)	IoU(%)	BIoU(%)
FCD-GAN [9]	09.05	85.86	16.38	08.92	21.15	30.91	70.25	42.93	27.33	13.49	18.78	52.59	27.68	16.06	15.82
TransWCD [13]	65.57	62.83	64.17	47.24	39.71	53.34	53.93	53.63	36.64	11.17	49.13	54.65	51.74	34.90	18.42
MJS-ICR-Net [17]	78.67	75.03	76.81	62.35	81.26	73.64	67.47	70.42	54.35	73.38	73.58	80.48	76.88	62.44	75.59
MS-Former [18]	80.75	57.32	67.05	50.43	74.48	62.36	52.31	56.90	39.76	59.81	74.51	82.84	78.44	64.53	66.73
ACWCD [12]	80.87	76.70	78.73	64.92	87.99	61.87	**85.37**	71.74	55.95	79.43	78.67	75.03	76.81	62.35	72.29
Ours	**89.56**	**94.92**	**92.16**	**85.46**	**95.31**	**84.82**	85.23	**85.02**	**73.94**	**85.36**	**86.91**	**90.84**	**88.83**	**79.93**	**89.22**

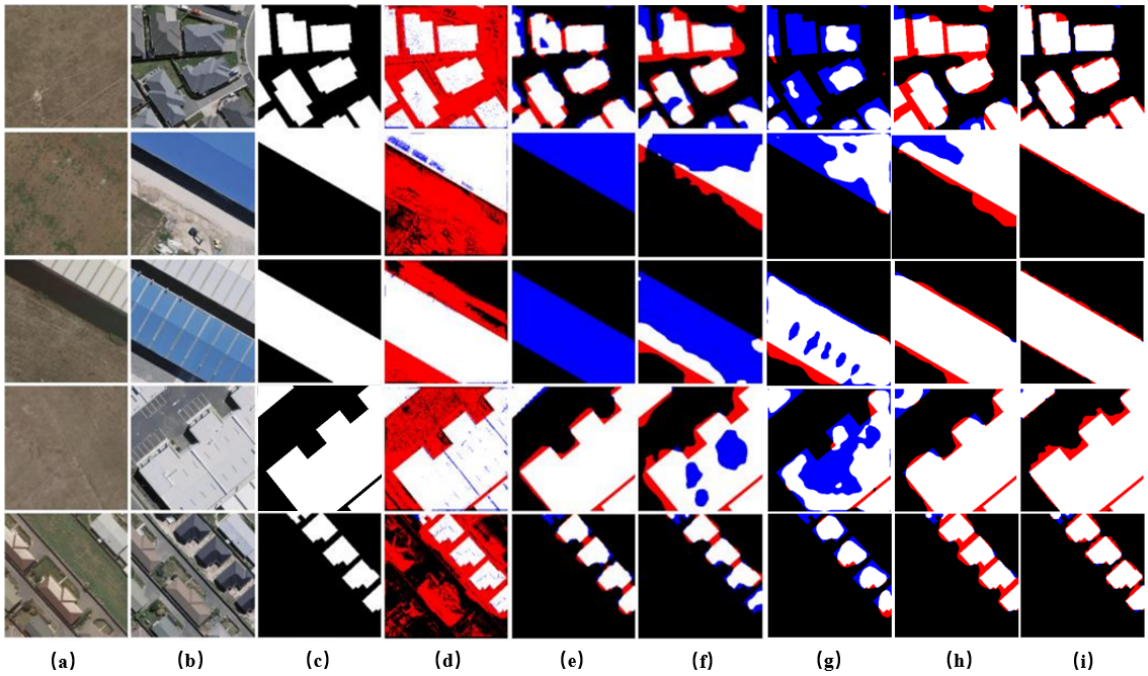


Fig. 5. **Visualization results on the WHU-CD dataset.** (a) Image I_1. (b) Image I_2. (c) Ground truth. (d) FCD-GAN. (e) TransWCD. (f) MJS-ICR-Net. (g) MS-Former. (h) ACWCD. (i) Our proposed method. TP, TN, FP, and FN are labeled in white, black, red, and blue.

outperforms the other weakly supervised approaches, achieving complete change localization, precise boundary delineation, and a substantial reduction in missed detections. On the contrary, FCD-GAN fails to reliably distinguish changed from unchanged regions, while MJS-ICR-Net and TransWCD exhibit noticeable omission errors; MS-Former demonstrates limited capacity in detecting small-scale changes, and ACWCD produces segmentation results with considerable boundary jaggedness. For large-scale changes, our model yields more complete predictions with fewer internal holes within the changed regions. Nevertheless, in scenes with densely adjacent structures, the model still faces difficulties in achieving clearly separated segmentations.

2) CD-DATA-GZ Dataset: As reported in Table II, our method attains the best performance on the CD-DATA-GZ dataset, achieving an $F1$-score of 85.02%, an IoU of 73.94%, and a BIoU of 85.36%. Relative to ACWCD ($F1$: 71.74%, IoU: 55.95%), our method improves $F1$ and IoU by 13.28% and 17.99%, respectively, and elevates the BIoU from 79.43% to 85.36%, reflecting superior boundary fidelity. Notably, although ACWCD attains the highest recall among the comparative methods (85.37%), its precision is limited to 61.87%, indicating a substantial number of false positives. On the contrary, our method achieves a comparable recall of 85.23% while raising the precision to 84.82%, thereby suppressing false positives without sacrificing detection completeness. Conversely, MS-Former suffers from an imbalanced precision–recall tradeoff, resulting in incomplete change localization, whereas TransWCD yields a markedly low BIoU of 11.17%, reflecting limited robustness to complex, large-scale changes.

Fig. 6 presents qualitative comparisons on the CD-DATA-GZ dataset, where five groups of large-scale building change samples are selected for evaluation. Our approach achieves superior performance by simultaneously reducing false positives and missed detections, delivering clear separation between neighboring large buildings and exhibiting robust behavior even in challenging scenarios. In particular, the proposed method accurately delineates the boundaries of large-scale changed regions with fewer internal

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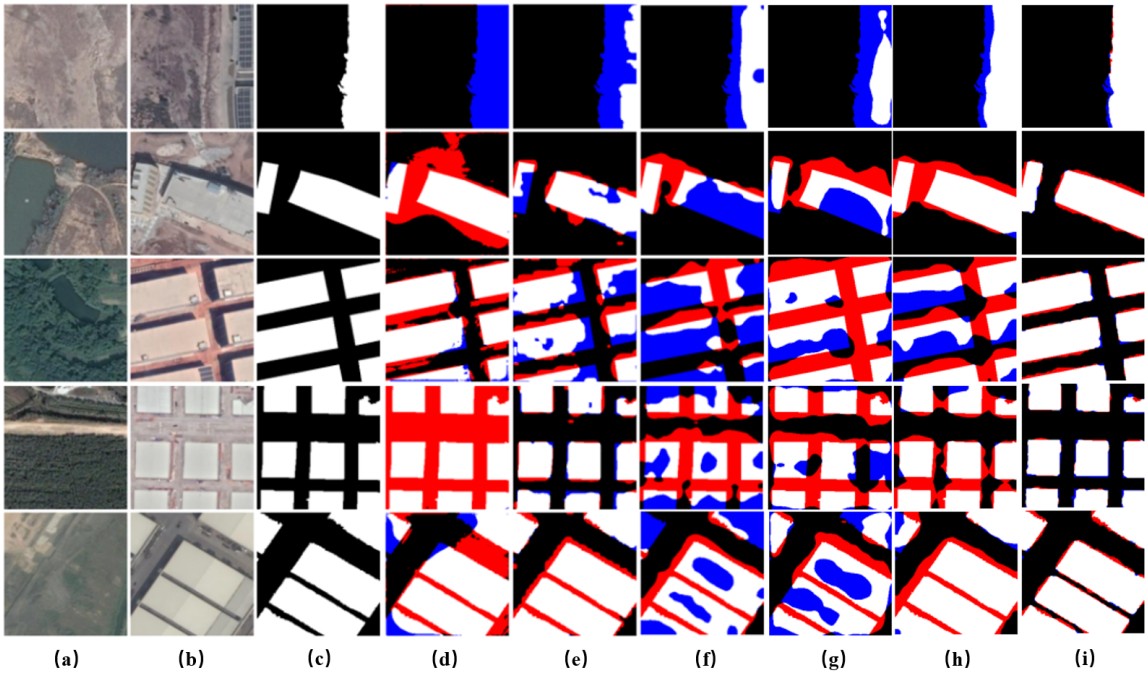


Fig. 6. **Visualization results on the CD-DATA-GZ dataset.** (a) Image I_1. (b) Image I_2. (c) Ground truth. (d) FCD-GAN. (e) TransWCD. (f) MJS-ICR-Net. (g) MS-Former. (h) ACWCD. (i) Our proposed method. TP, TN, FP, and FN are labeled in white, black, red, and blue.

holes, whereas the comparative methods tend to produce incomplete activations or blurred boundaries in the presence of complex backgrounds.

3) LEVIR-CD Dataset: As reported in Table II, our method attains the best performance on the LEVIR-CD dataset across all evaluation metrics, achieving an $F1$-score of 88.83%, an IoU of 79.93%, and a BIoU of 89.22%. Relative to MS-Former ($F1$: 78.44%, IoU: 64.53%), our method improves $F1$ and IoU by 10.39% and 15.40%, respectively, and elevates the BIoU from 75.59% (MJS-ICR-Net) to 89.22%, reflecting superior boundary fidelity. Moreover, our method attains both the highest precision (86.91%) and the highest recall (90.84%) among all approaches, demonstrating a well-balanced capability in detecting fine-grained building changes. On the contrary, ACWCD and MJS-ICR-Net achieve comparable $F1$-scores (76.81% and 76.88%, respectively) but with notably lower IoU, whereas FCD-GAN and TransWCD perform significantly worse, with $F1$-scores of 27.68% and 51.74%, respectively, confirming their limited ability to localize and segment change instances in large-scale, high-resolution scenes.

Fig. 7 presents qualitative comparisons involving changes in small-scale building clusters on the LEVIR-CD dataset. Compared with the other weakly supervised methods, our model produces more accurate boundaries for the change instances. Even in scenarios where the spectral differences between the bitemporal images are subtle, the proposed method is still capable of extracting relatively complete change features, thereby reducing missed detections and yielding clearer instance delineation.

D. Ablation Study

To validate the effectiveness of the proposed methods, we conducted a series of ablation experiments by progressively integrating each component into the baseline network and evaluating the corresponding performance.

1) Vision–Language Joint Learning Ablation Study: The investigated stages include the textual prompt, VLJR, and VLCA. As BoxLevelSet [40] operates on a single image input, we first extend it with a Siamese backbone to construct a baseline network suitable for CD. Starting from this baseline, we first incorporate the semantic description generation stage to inject text-guided features, yielding the Baseline+Text variant; this stage isolates the effect of textual prompt embedding by removing the text encoder and thereby eliminating language guidance from the network. Subsequently, we integrate the vision–language joint learning stage, decomposed into VLJR and VLCA, to progressively enhance the fusion of visual and textual features, producing the Baseline+Text+VLJR and Baseline+Text+VLJR+VLCA variants, respectively.

As shown in Table III, each module contributes positively to the final performance. The textual prompt alone improves the $F1$-score from 82.64% to 88.22% on WHU-CD, from 71.85% to 81.52% on CD-DATA-GZ, and from 80.44% to 85.78% on LEVIR-CD, indicating that the semantic descriptions enhance the discriminative capability of the visual features. Incorporating VLJR further raises the $F1$-score to 90.01%, 83.70%, and 87.01% on the three datasets, demonstrating the benefit of refining textual representations under visual guidance. Finally, adding VLCA yields the complete model, which attains $F1$-scores of 92.16%, 85.02%, and 88.83% and IoU values of 85.46%, 73.94%, and 79.93% on WHU-CD, CD-DATA-GZ, and LEVIR-CD. Compared with the baseline, the full model improves the $F1$-score by 9.52%, 13.17%, and 8.39% and the IoU by 15.03%, 17.87%, and 12.65% on the three datasets, respectively. These results confirm that every component of the

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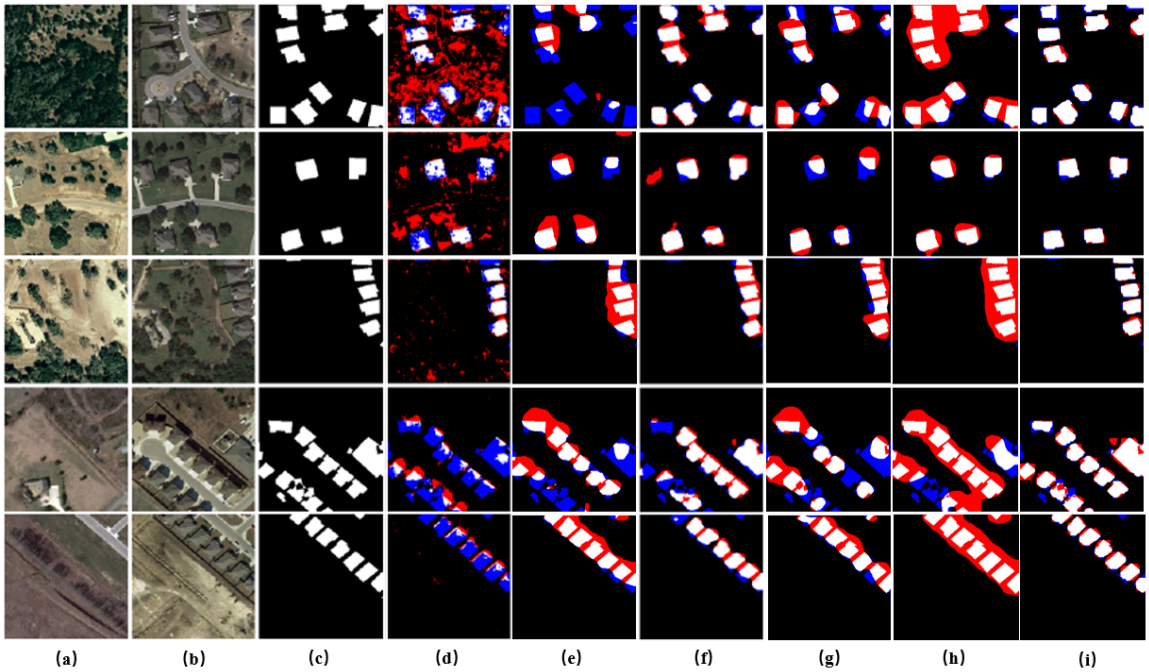


Fig. 7. **Visualization results on the LEVIR-CD dataset.** (a) Image I_1. (b) Image I_2. (c) Ground truth. (d) FCD-GAN. (e) TransWCD. (f) MJS-ICR-Net. (g) MS-Former. (h) ACWCD. (i) Our proposed method. TP, TN, FP, and FN are labeled in white, black, red, and blue.

TABLE III

ABLATION STUDY OF VISION–LANGUAGE JOINT LEARNING ON BENCHMARK DATASETS

Methods	WHU-CD				CD-DATA-GZ				LEVIR-CD			
	Pre.(%)	Rec.(%)	F1(%)	IoU(%)	Pre.(%)	Rec.(%)	F1(%)	IoU(%)	Pre.(%)	Rec.(%)	F1(%)	IoU(%)
Base	76.68	89.62	82.64	70.43	68.39	75.71	71.85	56.07	77.12	84.06	80.44	67.28
Base+Text	83.28	93.78	88.22	78.92	78.24	85.10	81.52	68.82	83.57	88.10	85.78	75.06
Base+Text+VLJR	85.27	**95.32**	90.01	81.82	82.67	84.77	83.70	71.97	84.80	89.33	87.01	77.00
Base+Text+VLJR+VLCA	**89.56**	94.92	**92.16**	**85.46**	**84.82**	**85.23**	**85.02**	**73.94**	**86.91**	**90.84**	**88.83**	**79.93**

TABLE IV

ABLATION STUDY OF ADAPTIVE BOUNDARY EVOLUTION ON BENCHMARK DATASETS

Methods	Modules		WHU-CD				CD-DATA-GZ				LEVIR-CD			
	SSCL	SSSL	Pre.(%)	Rec.(%)	F1(%)	IoU(%)	Pre.(%)	Rec.(%)	F1(%)	IoU(%)	Pre.(%)	Rec.(%)	F1(%)	IoU(%)
Network1	-	-	64.76	68.43	66.55	49.86	70.18	62.44	66.09	49.35	72.61	64.27	69.07	52.75
Network2	-	-	87.05	94.27	90.52	82.67	83.73	82.69	83.20	71.23	85.36	88.81	87.05	77.07
Network3	✓	-	88.58	93.98	91.20	83.83	83.99	83.75	83.87	72.22	**87.02**	89.31	88.15	78.81
Network4	-	✓	88.05	93.75	90.81	83.17	84.06	83.72	83.89	72.25	86.10	89.41	87.72	78.13
Network5	✓	✓	**89.56**	**94.92**	**92.16**	**85.46**	**84.82**	**85.23**	**85.02**	**73.94**	86.91	**90.84**	**88.83**	**79.93**

vision–language joint learning stage contributes positively, with the VLCA yielding the most pronounced gains.

2) Adaptive Boundary Evolution Module Ablation Study: The CIS module serves as the core component that enables detection under box-level supervision. To validate the advantage of box-level supervision over image-level supervision, we first construct a variant, denoted Network1, that localizes changes in an image-level (CAM-based) manner following GradCAM++ [47]; the comparison between Network1 and Network2 therefore isolates the effect of the supervision signal. Building on the box-supervised configuration, we then disentangle the joint self-supervised learning strategy into its two constituent constraints, self-supervised consistency learning (SSCL) and SSSL, and obtain the variants Network2 to Network5 by progressively incorporating them. In particular, Network2 applies box supervision without either constraint, Network3 and Network4 add SSCL and SSSL individually, respectively, and Network5 combines both, which corresponds to our complete model.

As shown in Table IV, Network1, which relies on CAM-based image-level supervision, attains $F1$-scores of only 66.55%, 66.09%, and 69.07% on WHU-CD, CD-DATA-GZ, and LEVIR-CD, respectively. Switching to box-level supervision (Network2) substantially improves the $F1$-score to 90.52%, 83.20%, and 87.05%, and raises the IoU from

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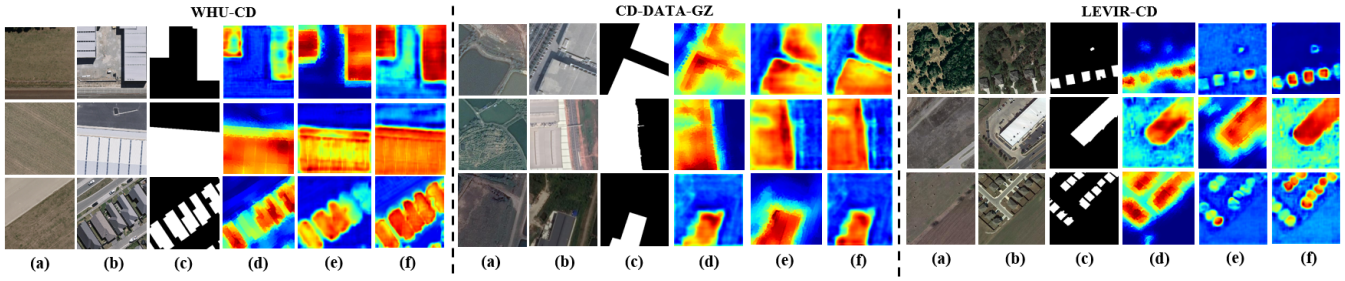


Fig. 8. **Heatmap visualization results on benchmark datasets.** (a) and (b) Initial images I_1 and I_2. (c) Ground truth. (d) Heatmap of this base method. (e) Heatmap of this Base+Text method. (f) Heatmap of our method. The red regions indicate higher activation values of the neural network, while the blue regions correspond to lower activation responses.

TABLE V
EVALUATION METRICS ON CD-DATA-GZ DATASET

Methods	Prec(%)	Rec(%)	F1(%)	IoU(%)
FC-EF [22]	86.44	81.02	83.65	71.89
SNUNet [1]	**90.10**	82.12	85.92	82.15
ChangeFormer [48]	86.89	84.44	85.65	75.09
BiT [25]	89.99	84.76	**87.30**	**85.52**
Ours	84.82	**85.23**	85.02	73.94

TABLE VI
EFFICIENCY EVALUATION RESULTS

Methods	Params(M)	FLOPs(G)	Time(ms)
Fully supervised change detection			
FC-EF [22]	1.67	4.13	234.79
SNUNet [1]	12.03	54.82	235.74
ChangeFormer [48]	41.02	202.62	245.41
BiT [25]	4.31	4.67	226.31
Weakly supervised change detection			
FCD-GAN [9]	43.35	161.78	246.67
TransWCD [13]	17.87	8.41	240.41
MJS-ICR-Net [17]	24.42	30.91	234.47
MS-Former [18]	15.19	10.33	236.33
ACWCD [12]	18.35	7.51	227.13
Ours	36.48	62.76	241.78

TABLE VII
PROMPT SELECTION STRATEGY ANALYSIS ON CD-DATA-GZ DATASET

Strategy	Prec(%)	Rec(%)	F1(%)	IoU(%)
Random	74.85	71.09	72.92	57.38
Only Building	80.71	79.19	79.94	66.58
Generate 3 categories	84.31	84.68	84.50	73.15
Generate 4 categories	**84.82**	85.23	**85.02**	**73.94**
Generate 5 categories	83.19	**85.47**	84.31	72.88

49.86%, 49.35%, and 52.75% to 82.67%, 71.23%, and 77.07%, respectively, confirming that the precise spatial information provided by box labels is essential for accurate change localization. When the joint self-supervised strategy is decomposed, adding SSCL alone (Network3) improves the $F1$-score on WHU-CD, CD-DATA-GZ, and LEVIR-CD (increase 0.68%, 0.67%, and 1.07%), whereas adding SSSL alone (Network4) raises the $F1$-score to 90.81%, 83.89%, and 87.72%. Combining both constraints (Network5) delivers the best results across all datasets, achieving $F1$-scores of 92.16%, 85.02%, and 88.83% and IoU values of 85.46%, 73.94%, and 79.93%, respectively, improving over the box-supervised baseline (Network2) by 1.64%, 1.82%, and 1.78% in $F1$ and by 2.79%, 2.71%, and 2.86% in IoU. These results indicate that the two self-supervised constraints are complementary: SSCL enhances boundary consistency, while SSSL strengthens feature similarity across scales, and their joint integration yields the most substantial gains.

3) Heatmaps Comparison: As shown in Fig. 8, we provide qualitative comparisons of the feature heatmaps produced by the proposed method and the baseline network across the three datasets. On the WHU-CD dataset, compared with the baseline, the introduction of the text modality yields more complete and coherent feature activation within large building instances. For regions containing small-scale building clusters, our model produces denser and more comprehensive activations, effectively reducing missed detections.

In the case of the LEVIR-CD dataset, the baseline heatmaps exhibit noticeable inconsistency in feature intensity within changed objects, reflecting weak internal responses. On the contrary, our method generates high-response regions that align more closely with the ground-truth masks.

On the CD-DATA-GZ dataset, the baseline fails to capture the structural integrity of changed instances, resulting in incomplete activation patterns. The text modality renders the activated regions more complete, and the proposed BoxCLIP framework further produces more coherent and object-aware attention maps, successfully delineating adjacent buildings with refined activation boundaries. Nevertheless, owing to the absence of precise pixel-level annotations in this dataset, underactivation may still occur in certain regions, and the predicted change masks occasionally present overly smooth or rounded edges.

Overall, these results demonstrate that each module contributes positively to the model's performance, highlighting their effectiveness and complementary benefits.

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TABLE VIII
HYPERPARAMETER ANALYSIS ON THREE BENCHMARK DATASETS

Hyper-parameters			WHU-CD				CD-DATA-GZ				LEVIR-CD			
λ_1	λ_2	λ_3	Prec(%)	Rec(%)	F1(%)	IoU(%)	Prec(%)	Rec(%)	F1(%)	IoU(%)	Prec(%)	Rec(%)	F1(%)	IoU(%)
1	1	1	89.43	93.11	91.22	83.86	84.62	85.11	84.86	73.70	86.59	90.80	88.65	79.61
1	2	1	89.65	**94.92**	**92.16**	**85.46**	84.82	85.23	**85.02**	**73.94**	86.82	90.77	88.75	79.76
2	1	1	**89.72**	94.21	91.90	85.01	**85.06**	84.89	84.97	73.87	86.29	90.50	88.34	79.13
1	2	2	89.58	94.32	91.88	84.98	84.50	**85.24**	84.87	73.71	**86.91**	**90.84**	**88.83**	**79.93**

V. DISCUSSION

A. Efficiency Evaluation With FSCD and WSCD Methods

As reported in Table V, we further conduct a direct comparison with four fully supervised CD methods on the CD-DATA-GZ dataset. Among these methods, FC-EF [22], SNUNet [1], ChangeFormer [48], and BiT [25] are trained using pixel-level annotations, whereas our model is trained under weakly supervised learning with only box-level labels. Despite the supervision gap, our method achieves highly competitive results.

To assess the computational efficiency and practical feasibility of our proposed method, we conducted a comparative analysis with several state-of-the-art CD methods. As shown in Table VI, we report the number of model parameters (Params), the number of floating-point operations (FLOPs), and the average inference time per sample (time).

These results demonstrate that our weakly supervised framework introduces only moderate parameter and computational overhead. Owing to the incorporation of the text modality, our model yields a larger parameter count than competing approaches such as ACWCD, while still incurring lower computational cost than FCD-GAN. Compared with FSCD, our method attains comparable detection accuracy while maintaining lower overhead than ChangeFormer. Overall, our method strikes a compelling balance between accuracy and efficiency, making it a promising candidate for practical CD tasks.

B. Prompt and Hyperparameter Analysis

1) Prompt Selection Strategy Analysis on CD-Data-GZ Dataset: To study the effect of the number of text embeddings, we vary this hyperparameter on the CD-DATA-GZ dataset, as shown in Table VII. When the category descriptions are selected at random, the text introduces excessive noise, yielding an $F1$-score of only 72.92% along with numerous false positives and missed detections. Restricting the prompts to the building category alone improves the $F1$-score by 7.02%. Nevertheless, because this strategy fails to incorporate other change instances that are present in the image, its detection accuracy remains lower than that achieved by the descriptions generated by the LLMs.

Increasing the number of embeddings from 3 to 4 improves the $F1$-score by 0.52% and the IoU by 0.79%. The best results are obtained with four text embeddings, yielding a precision of 84.82%, a recall of 85.23%, an $F1$-score of 85.02%, and an IoU of 73.94%. However, further increasing the number to five degrades the $F1$-score to 84.31%, accompanied by a sharp drop in precision to 83.19% despite the highest recall of 85.47%. This indicates that a moderate number of text embeddings provides sufficient category-level semantic guidance, whereas an excessive number introduces redundant or conflicting cues that weaken discriminative capability and cause more false positives.

2) Hyperparameter Analysis on Benchmark Datasets: Given that real-world remote sensing datasets vary significantly in terms of object scale, scene structure, noise levels, and annotation uncertainty, the optimal configuration of loss weights λ_1, λ_2, and λ_3 should be dynamically adapted to the characteristics of the data. Compared to using fixed weights across different scenarios, a more flexible and data-driven adjustment strategy better leverages the complementary strengths of each loss component. Selecting the highest $F1$ as the optimal result is the most robust and commonly used single optimality criterion in CD.

As shown in Table VIII, the optimal configuration is dataset-dependent and aligns with each benchmark's characteristics: WHU-CD, dominated by large-scale changes, favors stronger energy and projection constraints ($\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 1$, $F1$: 92.16%); CD-DATA-GZ, which exhibits complex and seasonally varying backgrounds, requires a balanced setting of ($\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 1$, $F1$: 85.02%), where overemphasizing any term degrades performance; LEVIR-CD, comprising fine-grained small building instances, benefits most from a stronger self-supervised boundary constraint ($\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 2$, $F1$: 88.83%). Overall, large-scale data favor structural/projection losses, small-scale data favor self-supervised boundary consistency, and noisy complex data require balanced weights.

VI. CONCLUSION

In this article, we have proposed BoxCLIP, a novel vision–language joint learning framework for box-supervised RSCD. By integrating multimodal supervision into a dual-branch architecture, our method effectively exploits both visual features and semantic textual cues to enhance detection performance under weakly supervised learning. Extensive experiments on three public datasets, WHU-CD, CD-DATA-GZ, and LEVIR-CD, demonstrate that BoxCLIP achieves competitive performance compared with existing weakly supervised methods.

Despite its competitive performance, the proposed method has three limitations. First, it has been validated only on BCD;

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gradual or amorphous land-cover transitions (e.g., water-body variations, deforestation, and agricultural dynamics) remain untested, as they fall outside the instance-level segmentation paradigm. Second, since CLIP is pretrained on natural image–text pairs rather than remote sensing data, a cross-domain gap may persist despite fine-tuning and frozen text encoders, potentially limiting semantic alignment. Third, the text modality increases the model size, which may hinder deployment on resource-limited platforms; more efficient multimodal interaction mechanisms remain an avenue for future work.

In future work, we aim to explore more fine-grained vision–language alignment strategies to better capture subtle semantic differences in CD while reducing the computational overhead. In addition, investigating multimodal prompt learning could facilitate the integration of diverse contextual cues, potentially enhancing the model's adaptability to varying scenes. Another promising direction is to extend the framework from discrete CD to continuous change perception, in which gradual changes over time need to be reliably identified. Furthermore, although BoxCLIP is currently designed for box-supervised CD, generalizing it to unsupervised settings would broaden its applicability and reduce annotation dependence.

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